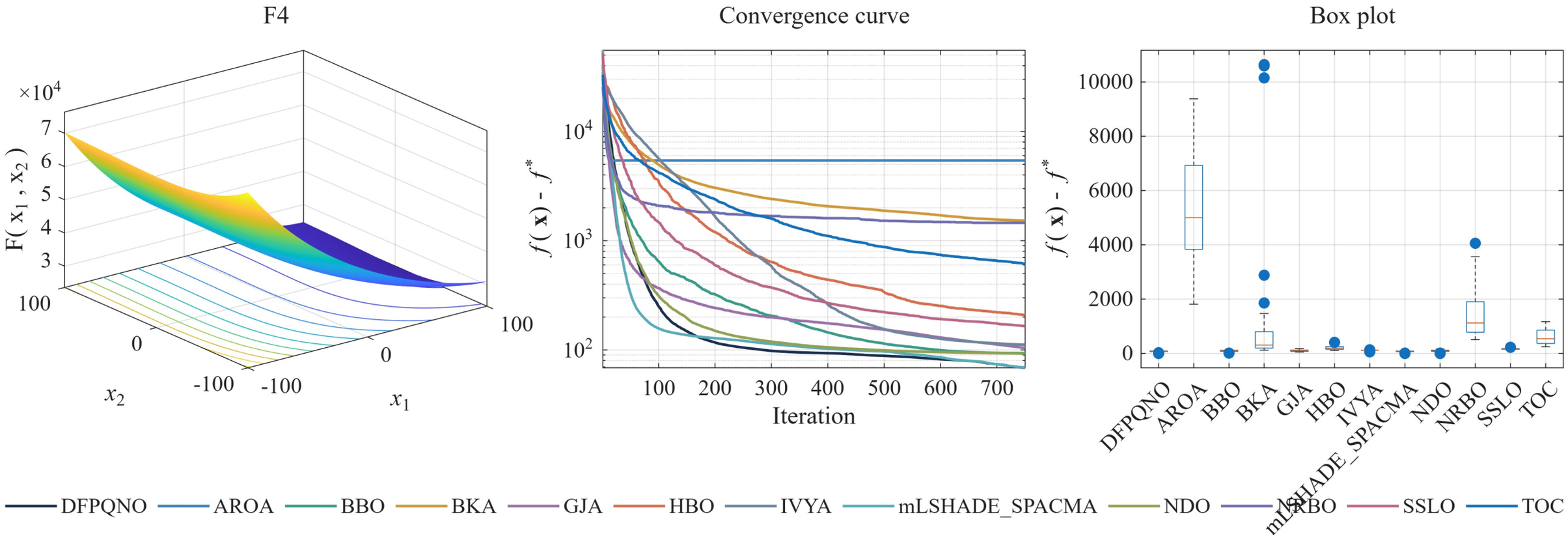

F4
×10^4
7
6
5
4
3
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
10^4
10^3
10^2
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
10000
8000
6000
4000
2000
0
f(x) - f*
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC